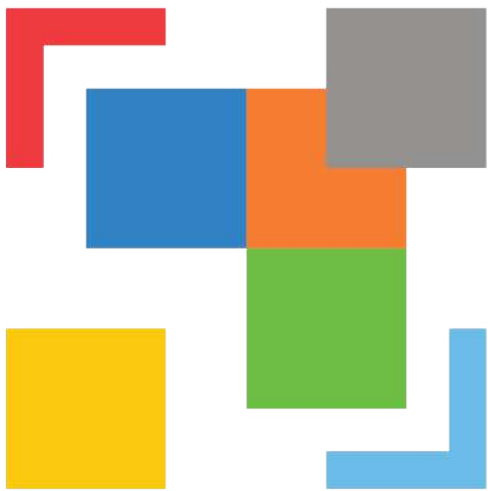




C.N.E

COMPUTER NETWORK
ENGINEERING



Al-Balqa Applied University
Faculty of Engineering Technology
Department of Scientific Basic Sciences
Midterm Exam
Applied mathematics for chemical engineers

Name: محمد عبد الوهاب

Date: 17/4/2022

Write the correct answer in the table (26 marks)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13
C	b	d	b	d	b	b	C	C	b	a	d	d

Q1) The differential equation $5x\left(\frac{dy}{dx}\right)^4 + 2x\left(\frac{d^3y}{dx^3}\right)^2 + x^2\left(\frac{d^2y}{dx^2}\right)^3 = 2e^x$

- a) linear differential equation of order 4 and degree 3
b) nonlinear differential equation of order 4 and degree 3
c) nonlinear differential equation of order 3 and degree 2
d) linear differential equation of order 3 and degree 4

Q2) The solution of the homogeneous differential equation $y'' - y' - 6y = 0$ is

- a) $y = c_1e^x + c_2e^{-6x}$
b) $y = c_1e^{3x} + c_2e^{-2x}$
c) $y = c_1e^{-3x} + c_2e^{2x}$
d) $y = c_1e^{-x} + c_2e^{6x}$

Q3) The solution of the d.e. $(xy^2 + 4x + 3y^2 + 12)dx = (xy + 4 + 2y + 2x)dy$ is

- a) $(x+3)\ln|x+2| = \frac{1}{2}\ln|y^2+4| + 2\tan^{-1}\frac{y}{2} + c$
b) $\frac{x^2}{2} + 3\ln|x+2| = \frac{1}{2}\ln|y-2| + c$
c) $\ln|x+2| + 3x = \frac{y^2}{2} - 2y + c$
d) $x + \ln|x+2| = \frac{1}{2}\ln|y^2+4| + \tan^{-1}\frac{y}{2} + c$

Q4) If the d.e. $(ye^{2xy} + x)dx + 2bxe^{2xy}dy = 0$ is exact D.E. then b=

- a) 1
b) $\frac{1}{2}$
c) 2
d) 4

Q5) The solution of the homogeneous d.e. $y'' - 2y' - y = 0$ is

- a) $y = c_1e^x + c_2e^{-2x}$
b) $y = c_1e^x \cos\sqrt{2}x + c_2e^x \sin\sqrt{2}x$
c) $y = c_1e^x \cos 2x + c_2e^x \sin 2x$
d) $y = c_1e^{(1+\sqrt{2})x} + c_2e^{(1-\sqrt{2})x}$

Q6) If $f(x, y) = c$ is a solution for the exact d.e. $2e^{2x} \cos y dx - e^{2x} \sin y dy = 0$, then $\frac{\partial f}{\partial x} =$

- a) $2e^{2x} \cos y$
b) $-2e^{2x} \sin y$
c) $-e^{2x} \cos y$
d) $-e^{2x} \sin y$

Q7) The integrating factor of the linear d.e. $\sin y \frac{dx}{dy} + x \cos y = 1$ is

- a) $\ln \sin y$ ☒ b) $\sin y$ c) $\cos y$ d) $\csc y$

Q8) $y = c_1 e^{-3x} \sin \sqrt{2}x + c_2 e^{-3x} \cos \sqrt{2}x$ is solution for the 2nd order d.e.

- a) $y'' - 6y' + 13y = 0$ b) $y'' + 6y' + 5y = 0$ ☒ c) $y'' + 6y' + 11y = 0$ d) $y'' - 6y' + 11y = 0$

Q9) One of the following is a homogeneous function

- a) $f(x, y) = 3x + \frac{y^2}{x} + \tan\left(\frac{x^2}{y}\right)$ b) $f(x, y) = x^2 + \sqrt{xy} + y^2 e^{\frac{x}{y}}$
☒ c) $f(x, y) = x^3 + xy^2 + y^3 \ln\left(\frac{x}{y}\right)$ d) $f(x, y) = y^4 + x^2 y^2 + \sin\left(\frac{x^2}{xy}\right)$

Q10) The general solution of the d.e. $y' = 1 - (x - y)^2$

- a) $y = ye^y + cx$ ☒ b) $\frac{1}{y-x} = x + c$ c) $\frac{(x-y)^3}{3} = x + c$ d) $\ln \left| \frac{1-y}{1+y} \right| = x + c$

Q11) The Initial value problem for the solution $y = \frac{\cos^2 x - \sin^2 x}{e^x}$, $\cos 2x e^{-x}$, $\alpha = -1$, $\beta = 2$

- ☒ a) $y'' + 2y' + 5y = 0$, $y(0) = 1$, $y'(0) = -1$ b) $y'' - 2y' + 5y = 0$, $y(0) = 1$, $y'(0) = -1$
c) $y'' + 2y' + 2y = 0$, $y(0) = y'(0) = 1$ d) $y'' - y' + 2y = 0$, $y(0) = y'(0) = 1$

Q12) The solution of the d.e. $\frac{dy}{dx} = \sqrt{\frac{x^2 - y^2}{x^2}} + \frac{y}{x}$ is

- a) $y = \tan^{-1}\left(\frac{y}{x}\right) + c$ b) $y = x \sin^{-1} \sqrt{1 - cx^2}$ c) $y = \ln(1 - x^2) + c$ ☒ d) $\sin^{-1}\left(\frac{y}{x}\right) = \ln x + c$

Q13) Transforming the 2nd order D.E. $yy'' - (y')^2 = 0$ to 1st order D.E. we get

- a) $(y')^2 = 2y + c$ b) $y' = \frac{1}{y+c}$ c) $y' = c \ln y$ ☒ d) $y' = cy$

Q14) (4 marks) solve the Bernoulli's differential equation $xy' - x^3y^3 = y$
(with all steps)

$$xy' - x^3y^3 = y \quad \div x$$

$$y' - x^2y^3 = \frac{1}{x}y \quad \text{ ~~$x=y$~~ }$$

~~$$y' - x^2y^3 = \frac{1}{x}y$$~~

$$y' - \frac{1}{x}y = x^2y^3$$

$$\div y^{-3}$$

$$y^{-3}y' - \frac{1}{x}y^{-2} = x^2$$

$$y^{-2} = u$$

$$\boxed{-2y^{-3}y' = u'}$$

$$\frac{u'}{-2} - \frac{1}{x}u = x^2$$

$$u' + \frac{2}{x}u = -2x^2$$

$$I(x) = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$$

$$u = \frac{1}{x^2} \left(\int x^2 \cdot -2x^2 dx + C \right)$$

$$u = \frac{1}{x^2} \left(-2 \int x^4 dx + C \right)$$

$$u = \frac{1}{x^2} \left(-2 \frac{x^5}{5} + C \right)$$

$$u = -\frac{2}{5}x^3 + Cx^{-2}$$

$$y^{-2} = -\frac{2}{5}x^3 + Cx^{-2}$$

$$y = \left(-\frac{2}{5}x^3 + Cx^{-2} \right)^{-\frac{1}{2}}$$

4

إختبار - فنونج المير لسنة 2022 ففيل نأني

Q2 $y'' - y' - 6y = 0$ $r^2 - r - 6 = 0$

$(r-3)(r+2) = 0$ $r = -2, 3$

$y = c_1 e^{-2x} + c_2 e^{3x}$ [b]

Q3 $(xy^2 + 4x + 3y^2 + 12) dx = (xy + 4 + 2y + 2x) dy$

$(x(y^2 + 4) + 3(y^2 + 4)) dx = (x(y+2) + 2(y+2)) dy$

$(y^2 + 4)(x+3) dx = (y+2)(x+2) dy$

$\frac{y+2}{y^2+4} dy = \frac{x+3}{x+2} dx$

$\int \frac{y}{y^2+4} + \frac{2}{y^2+4} dy = \int \frac{x}{x+2} + \frac{3}{x+2} dx$

$\frac{1}{2} \ln|y^2+4| + \frac{2}{\frac{y}{2}} \tan^{-1}\left(\frac{y}{2}\right) = x+2 - 2 \ln|x+2| + 3 \ln|x+2| + C$

$\frac{1}{2} \ln|y^2+4| + \tan^{-1}\left(\frac{y}{2}\right) = x+2 + \ln|x+2| + C$

$x+2 = z$ $x = z-2$

$\frac{1}{2} \ln|y^2+4| + \tan^{-1}\left(\frac{y}{2}\right) = x + \ln|x+2| + C$

$dz = dx$

$\int \frac{z-2}{z} dz = \int 1 - \frac{2}{z} dz$

$z - 2 \ln z$

$x+2 - 2 \ln|x+2|$

[d]



Q4) $(ye^{2xy} + x)dx + 2bx e^{2xy} dy = 0$

$M \quad dx \quad N \quad dy$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$2yx e^{2xy} + e^{2xy} = 4xy e^{2xy} + 2b e^{2xy} \quad \div e^{2xy}$$

$$2yx + 1 = 4xyb + 2b$$

قيمة b التي تجعل
المعادلة متعادلة

$$\boxed{\frac{1}{2}}$$

$$\boxed{b}$$

5) $y'' - 2y' - y = 0$

$$\Delta = 4 - 4 \times (-1) = 8 \quad \checkmark$$

$$r^2 - 2r - 1 = 0$$

$$r = \frac{2 \pm \sqrt{8}}{2} = \frac{1 \pm \sqrt{2}}{1}$$

$$y = C_1 e^{(1+\sqrt{2})x} + C_2 e^{(1-\sqrt{2})x}$$

$$\boxed{d}$$

Q7) $\sin y \frac{dx}{dy} + x \cos y = 1 \quad \div \sin y \quad \text{linear in } x$

$$\frac{dx}{dy} + \cot y x = 1$$

$$I(y) = e^{\int \cot y dy} = e^{\int \frac{\cos y}{\sin y} dy} = e^{\ln |\sin y|} = \sin y$$

$$\boxed{b}$$



$$[8] \quad y = C_1 e^{-3x} \sin \sqrt{2}x + C_2 e^{-3x} \cos \sqrt{2}x$$

$$\alpha = -3 \quad \beta = \sqrt{2}$$

$$(r - (-3))^2 + (\sqrt{2})^2$$

$$r^2 + 6r + 9 + 2 \rightarrow r^2 + 6r + 11$$

$$y'' + 6y' + 11y = 0$$

C

[9]

نقطة مكان $x \leftarrow x$

$y \leftarrow y$

نأخذها على مشترك ليعود الاقتران الأصلي

هذه السلسلة متطابقة فقط على

$$P(x, y) = x^3 x^3 + x^3 x y^2 + x^3 y^3 \ln\left(\frac{x}{y}\right)$$

$$x^3 (x^3 + x y^2 + y^3 \ln \frac{x}{y})$$

Homogeneous

C

[10]

$$y' = 1 - (x - y)^2$$

$$x - y = z$$

$$1 - dy = dz \rightarrow \boxed{dy = 1 - dz}$$

$$x \cdot dz = x \cdot u^2$$

$$\frac{dz}{dx} = u^2 \rightarrow \frac{1}{u^2} du = dx$$

$$\frac{u^{-1}}{-1} = x + C \rightarrow \left(\frac{-1}{x - y} \right) = x + C = \frac{1}{y - x} = x + C \quad [b]$$



III $\frac{\cos^2 x - \sin^2 x}{e^x}$ متطابقة

$$\frac{\cos 2x}{e^x} = e^{-x} \cos 2x \quad \alpha = -1 \quad \beta = 2$$

$$(r+1)^2 + 4 \rightarrow r^2 + 2r + 5$$

$$y'' + 2y' + 5y = 0 \quad \checkmark \quad \boxed{A}$$

$$\boxed{12} \quad \frac{dy}{dx} = \sqrt{\frac{x^2 - y^2}{x^2}} + \frac{y}{x} \quad \frac{y}{x} = u$$

$$\rightarrow y = xu$$

$$\frac{xdu + udx}{dx} = \sqrt{\frac{x^2 - x^2u^2}{x^2}} + \frac{xu}{x} \rightarrow dy = xdu + udx$$

$$\frac{xdu + udx}{dx} = \sqrt{1 - u^2} + u \rightarrow xdu + udx = \sqrt{1 - u^2} dx + udx$$

$$xdu = \sqrt{1 - u^2} dx \Rightarrow \int \frac{1}{x} dx = \int \frac{1}{\sqrt{1 - u^2}} du$$

$$\ln |x| + C = \sin^{-1}(u) \rightarrow \sin^{-1}\left(\frac{y}{x}\right) = \ln |x| + C \quad \boxed{d}$$

$$\boxed{13} \quad yy'' - (y')^2 = 0 \quad y' = u \quad y'' = uu'$$

$$yuu' - u^2 = 0 \rightarrow \frac{u}{u^2} du = \frac{1}{y} dy \Rightarrow \ln u = \ln y + C$$

$$u = C \rightarrow y' = yC \quad \boxed{d}$$





كبير على موضوع
الكوشي أويلر

Al-Balqa Applied University
Faculty of Engineering Technology
Department of Scientific Basic Sciences
Applied mathematics for chemical engineers
Quiz1

10

قسم هندسة

Name: Maysam Salqahat

Date: 17/5/2022

Write the correct answer in the table

Q1	Q2	Q3	Q4	Q5
b	d	b	d	a

Q1) The solution of the Cauchy homogeneous differential equation $x^2 y'' + 3xy' + 3y = 0$

a) $y = \frac{c_1 \sin 2 \ln x + c_2 \cos 2 \ln x}{x}$

b) $y = \frac{c_1 \sin \sqrt{2} \ln x + c_2 \cos \sqrt{2} \ln x}{x}$

c) $y = c_1 \frac{\sin \sqrt{2} x}{x} + c_2 \frac{\cos \sqrt{2} x}{x}$

d) $y = c_1 e^{-x} \sin \sqrt{2} x + c_2 e^{-x} \cos \sqrt{2} x$

$1 - 4 \times 3 = 4 - 12$

$m(m-1) + 3m + 3 = 0$
 $m^2 + 2m + 3 = 0$
 $m = \frac{-2 \pm \sqrt{4 - 12}}{2}$

Q2) The solution of the Cauchy homogeneous differential equation $y'' - \frac{2}{x} y' - \frac{4}{x^2} y = 0$ is

a) $y = c_1 x^{-4} + c_2 x^{-1}$

b) $y = c_1 x^{-2} + c_2 x$

c) $y = c_1 x^{-2} + c_2 x^2$

d) $y = c_1 x^4 + c_2 x^{-1}$

$m^2 - 3m - 4 = 0$
 $m = \frac{3 \pm \sqrt{9 + 16}}{2}$
 $m = \frac{3 \pm 5}{2}$
 $m = 4, -1$

Q3) The solution of the Cauchy homogeneous differential equation $x^2 y'' - xy' + y = 0$ is

a) $y = c_1 x^{-1} + c_2 x$

b) $y = c_1 x + c_2 x \ln x$

c) $y = c_1 x + c_2 x e^x$

d) $y = c_1 x^{-2} + c_2 x$

Q4) The solution of a homogeneous Cauchy d.e. is $y = c_1 x^3 + c_2 x^3 \ln x$.

Then the Cauchy d.e. will be

a) $x^2 y'' - xy' + 2y = 0$

b) $x^2 y'' - 6xy' + 9y = 0$

c) $y'' - 3y' + 9y = 0$

d) $x^2 y'' - 5xy' + 9y = 0$

$m(m-3)(m-3) = 0$
 $m^2 - 6m + 9 = 0$
 $(m-3)^2 = 0$
 $m = 3$
 $x^2 y'' - 5xy' + 9y = 0$

Q5) The characteristic equation of the Cauchy d.e $3x^2 y'' - xy' + 2y = 0$ is

a) $3m^2 - 4m + 2 = 0$

b) $3m^2 - 3m + 2 = 0$

c) $3m^2 - m + 2 = 0$

d) $3m^2 - 3m = 0$

$3(m(m-1) - m + 2)$